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STA510 – Assignment 2 report

1) b) $f(t) = \frac{e^{-t/2}}{1+t^2}$

$$f'(t) = \frac{-\frac{1}{2}e^{-t/2}(t^2+4t+1)}{(1+t^2)^2} = -\frac{e^{-t/2}(t^2+4t+1)}{2(1+t^2)^2}$$

$$\forall t \in [0,2]: \begin{cases} e^{-t/2} > 0 \\ t^2+4t+1 > 0 \\ 2(1+t^2)^2 > 0 \end{cases}$$

THEREFORE $f'(t) < 0 \quad \forall t \in [0,2]$

e) $G(t) = \int_0^t g(s)ds = \frac{1}{2(1-e^{-1})} \int_0^t e^{-s/2} ds$

$$= \frac{[-2e^{-s/2}]_0^t}{2(1-e^{-1})} = \frac{1-e^{-t/2}}{1-e^{-1}}, \quad 0 \le t \le 2$$

$$u = \frac{1-e^{-t/2}}{1-e^{-1}} \rightarrow u(1-e^{-1}) = 1-e^{-t/2}$$

$$-\frac{t}{2} = \log(1-u(1-e^{-1})) \rightarrow G^{-1}(u) = -2\log(1-u(1-e^{-1}))$$

PROBLEM 2

a)

$$\alpha = \frac{1}{\lambda^k} = \frac{1}{6^2} = \frac{1}{36} \quad ; \quad \beta = k = 2$$

$$E(T) = \alpha^{-\frac{1}{\beta}} \Gamma\left(1 + \frac{1}{\beta}\right) = \left(\frac{1}{36}\right)^{-\frac{1}{2}} \Gamma\left(\frac{3}{2}\right)$$

$$= 6 \cdot \frac{\sqrt{\pi}}{2} = 3\sqrt{\pi} = \boxed{5.317}$$

$$Var(T) = \alpha^{-\frac{2}{\beta}} \left[\Gamma\left(1 + \frac{2}{\beta}\right) - \left(\Gamma\left(1 + \frac{1}{\beta}\right)\right)^2\right]$$

$$= 36 \left[\Gamma(2) - \left(\Gamma\left(\frac{3}{2}\right)\right)^2\right]$$

$$= 36 \left[1 - \left(\frac{\sqrt{\pi}}{2}\right)^2\right] = 36\left(1 - \frac{\pi}{4}\right) = 9(4 - \pi)$$

$$SD = \sqrt{9(4-\pi)} = 3\sqrt{4-\pi} = \boxed{2.779}$$

d) $$\Lambda(t) = \int_0^t \lambda(u)\,du = \int_0^t \left[3 + 2\cos\left(\frac{\pi u}{6}\right)\right] du$$

$$= \left[3u + \frac{12}{\pi}\sin\left(\frac{\pi u}{6}\right)\right]_0^t = \boxed{3t + \frac{12}{\pi}\sin\left(\frac{\pi t}{6}\right)}$$

$$E(N(36)) = \Lambda(36) = 3 \cdot 36 + \frac{12}{\pi}\sin\left(\frac{\pi \cdot 36}{6}\right) = 108 + \frac{12}{\pi}\sin(6\pi)$$

$$\stackrel{!}{=} \boxed{108} \text{ FAILURES OVER 3 YEARS}$$

$$N(36) \sim \text{Poisson}(108)$$